Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3.

Multilingual support for template handling
Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation
Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types
Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers
TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

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listings,textcomp,color latexsym,xcolor,multicol,booktabs,calligra

amsmath,amssymb,BOONDOX-cal,bm

graphicx,pstricks,stackengine

webmedia

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Exemplo de Título

Template de Apresentações do WebMedia 2022

Fulana de Tal1[1] Ciclano de Tal2[2]

[1]Departamento de Informática
Universidade Tecnológica Federal do Paraná

[2]Departamento de Matemática
Universidade Federal de XX

11 de Novembro de 2022

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Sumário

Para sumário automático, inclua seções.

Introdução

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Objetivos

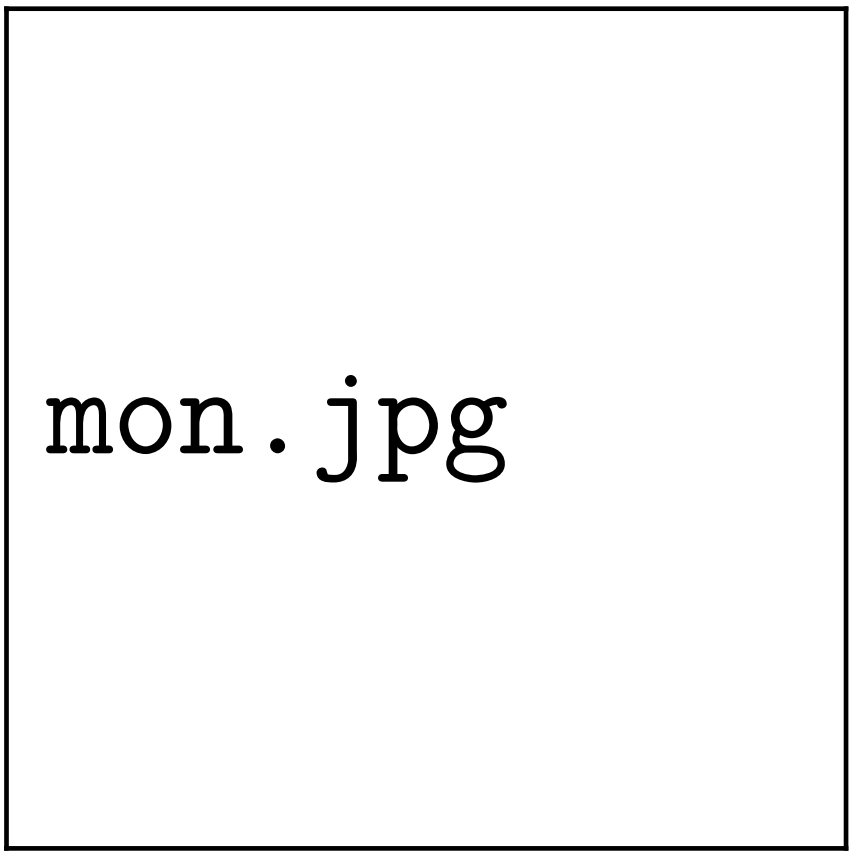
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Metodologia

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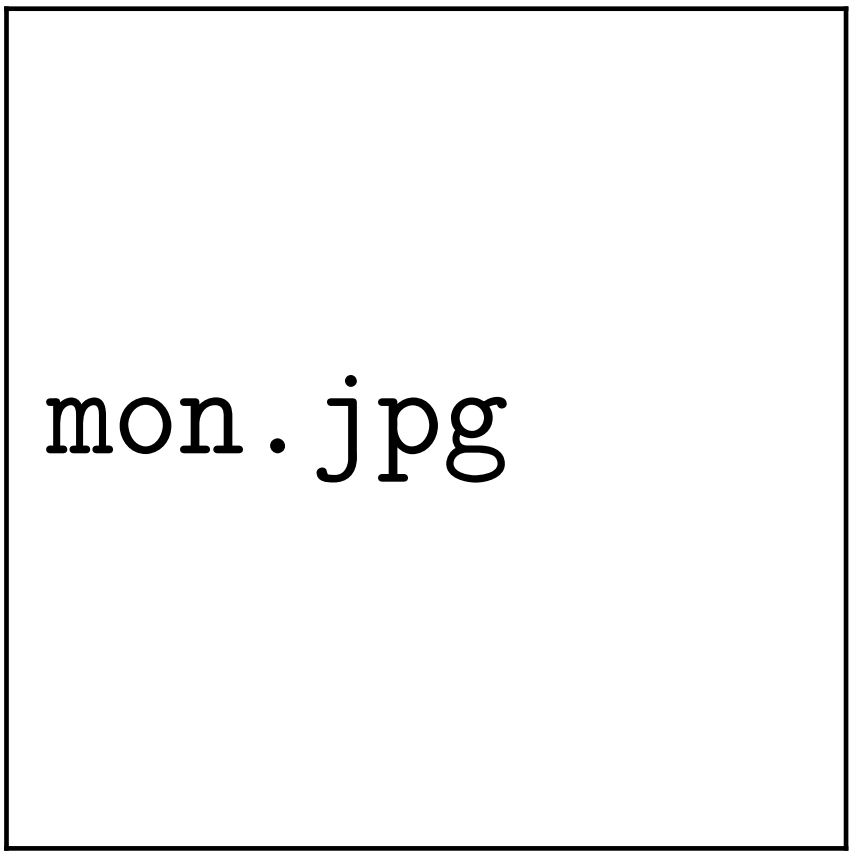
Metodologia



Gambar: Test

- Blah blah blah
  - Blah blah blah

Metodologia



Gambar: Test

- Blah blah blah
  - Blah blah blah
- Blah blah blah
  - Blah blah blah

Exemplo de código

Exemplo de código em Python.
[language=Python,keywordstyle=,**stringstyle**=] *Fibonacci numbers module*
def fib(n): write Fibonacci series up to n a, b = 0, 1 while a < n: print(a, end=' ') a, b = b, a+b print()

Exemplo de bloco

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Resultados

- First item
- Second item
- Third item



Resultados

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	Mean	Sd. Dev.	95% HPD
μ_1	1.220	0.303	[0.567, 1.821]

Resultados

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	Mean	Sd. Dev.	95% HPD
μ_1	1.220	0.303	[0.567, 1.821]
μ_2	2.676	0.409	[1.863, 3.498]

Resultados

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	Mean	Sd. Dev.	95% HPD
μ_1	1.220	0.303	[0.567, 1.821]
μ_2	2.676	0.409	[1.863, 3.498]
ρ	0.313	0.264	[-0.295, 0.749]

Conclusões

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Bibliography I



Peter Tankov.
Financial Modelling with Jump Processes.
Editora Taylor & Francis , ano 2003.



Sato, K.
Levy Processes and Infinitely Divisible Distributions.
Editora Cambridge University Press , ano 1999.



Protter, P.
Stochastic Integration and Differential Equations.
U.S. Government Printing Office , ano 2004.